10/10/2025

# Tutorial

Discrete time signal $x[n]$

① $x[n]$ is periodic with period $N = 6$ $\Rightarrow$ $\omega_0 = \frac{2\pi}{N} = \frac{2\pi}{6}$

$\omega_0 = \pi/3$

② $\sum_{n=0}^{5} x[n] = 2$

$$a_k = \frac{1}{N} \sum_{n=<N>} x[n] e^{-jk\frac{2\pi}{N}n}$$

$$a_0 = \frac{1}{N} \sum_{n=<N>} x[n]$$

$\frac{1}{6} \sum_{n=0}^{5} x[n] = a_0 = \frac{2}{6} = 1/3 = a_0$

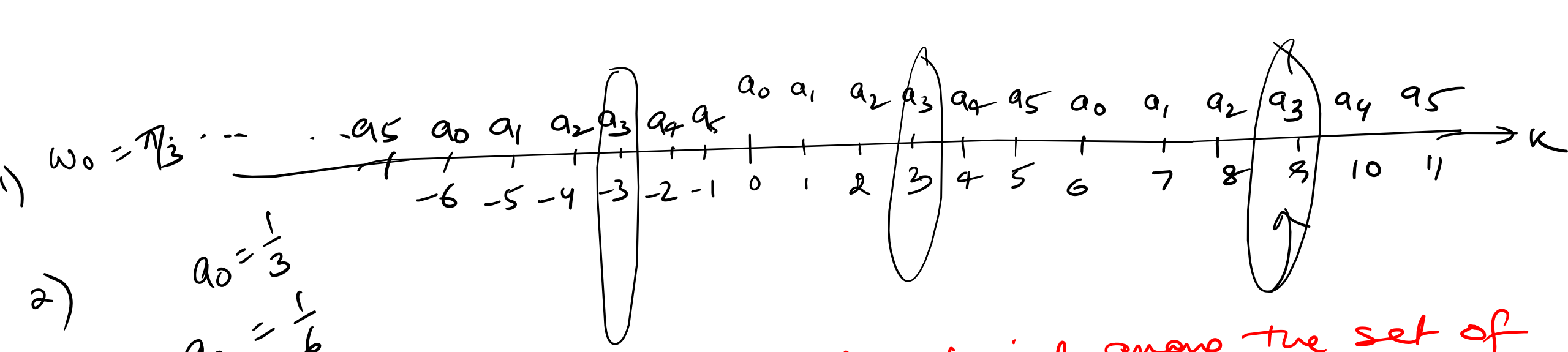
③ $\sum_{n=2}^{7} (-1)^n x[n] = 1$ $\Rightarrow$ $\frac{1}{N} \sum_{n=2} \left(e^{-j\pi n}\right) x[n] = \frac{1}{N}$

$e^{-j3\omega_0 n} = e^{-jk\frac{2\pi}{N}n}$ $\Rightarrow$ $k\frac{2}{N} = 1$

$k = 3$

$\frac{1}{6} \sum_{n=<N>} e^{-j3\omega_0 n} = \frac{1}{6}$

$a_3 = 1/6$




1)

2) $a_0 = \frac{1}{3}$

$a_3 = \frac{1}{6}$

3) ④ $x[n]$ has min. power per period among the set of signals satisfying the above three conditions.

((

$$\frac{1}{N} \sum_{n=<N>} |x[n]|^2 = \sum_{k=<N>} |a_k|^2$$

$\Rightarrow$ $a_1 = a_2 = a_4 = a_5 = 0$

$$x[n] = \frac{1}{3} + \frac{1}{6} e^{j\frac{3.2\pi}{6}n} = \frac{1}{3} + \frac{1}{6} e^{j\pi n}$$

3.41

C.T. periodic signal with period 3 $\Rightarrow T = 3$

& F.S. coefficients —

$$\omega_0 = \frac{2\pi}{3}$$

$$x(t) = \sum_{k=0}^{\infty} a_k e^{jk\omega_0 t}$$

$$a_k = \frac{1}{T}\int_T x(t) e^{-jk\omega_0 t}\, dt$$

$$a_0 = \frac{1}{T}\int_T x(t)\, dt$$

(a) $a_k = a_{k+2}$

$\Rightarrow$ $x(t)$ is non-zero at integral multiples of $t = 3/2$. $\Rightarrow$ $\delta(t)$'s

(b) $a_k = a_{-k}$ $x(t) = x(-t)$ ✓

$\delta(t)$ is an even fn. ✓

(c) $\int_{-0.5}^{0.5} x(t)\, dt = 1$

(d) $\int_{1}^{2} x(t)\, dt = 2$

$a_0 = a_2 = a_4 = \cdots$

$a_1 = a_3 = a_5 = \cdots$

(a) $a_{k+2} = a_k$ →

$$\frac{1}{T}\int_T x(t)\, e^{-j(k+2)\omega_0 t}\, dt = \frac{1}{T}\int_T x(t)\, e^{-jk\omega_0 t}\, dt$$

$$\Rightarrow \frac{1}{T}\int_T x(t)\, e^{-jk\omega_0 t}\, e^{-j2\omega_0 t}\, dt = \frac{1}{T}\int_T x(t)\, e^{-jk\omega_0 t}\, dt$$

$$\Rightarrow \frac{1}{T}\int_T \left(x(t)\, e^{-j2\omega_0 t}\right) e^{-jk\omega_0 t}\, dt = \frac{1}{T}\int_T x(t)\, e^{-jk\omega_0 t}\, dt$$

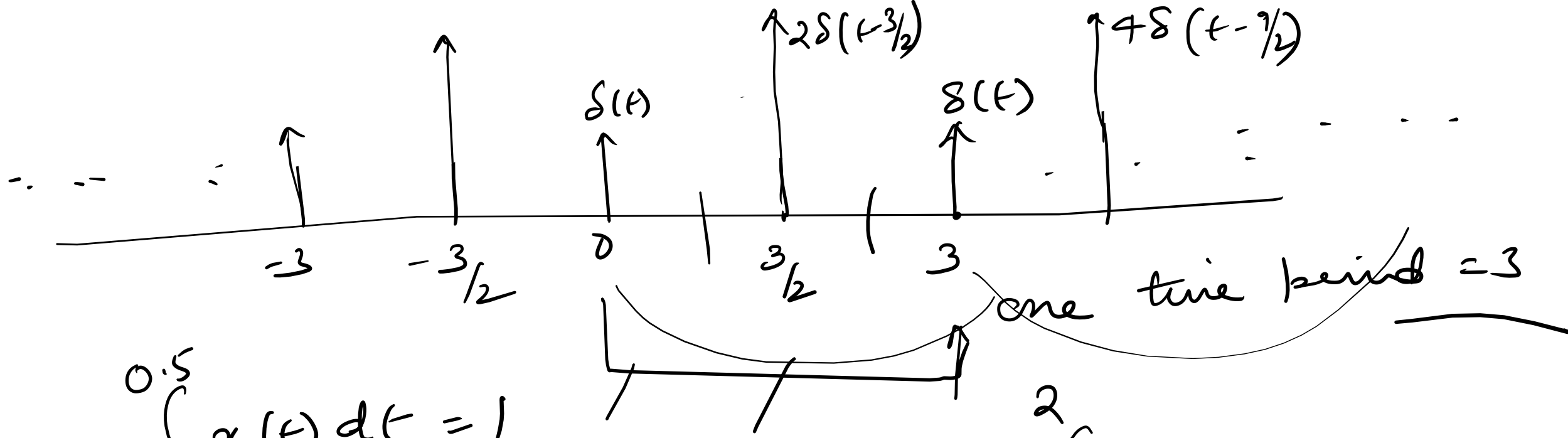
$$\Rightarrow x(t) = x(t)\, e^{-j2\omega_0 t}$$

$$\Rightarrow \boxed{x(t) = x(t)\, e^{-j2.\frac{2\pi}{3} t}}$$

$x(t)$ is non-zero at $t = 0, t = 3/2, t = 3, t = 9/2 \cdots$

$2.\frac{2\pi}{3} t = l.2\pi$

$t = l.\frac{3}{2}$




$$\int_{-0.5}^{0.5} x(t)\,dt = 1$$

$$\int_{-0.5}^{0.5} \delta(t)\,dt = 1$$

$$\int_{1}^{2} x(t)\,dt = 2$$

$$\int_{1}^{2} a\,\delta(t-3/2)\,dt = 2$$

$$a = 2$$

$$x(t) = \sum_{k=-\infty}^{\infty} \delta(t-3k) + 2\sum_{k=-\infty}^{\infty} \delta\left(t-3k-\tfrac{3}{2}\right)$$

3.44

① $x(t)$ is real signal.

② $x(t)$ is periodic with period $T=6$ & has F.S. $a_k$'s.

③ $a_k = 0$ for $k=0$ & $k>2$

$a_k = a_{-k}^* \Leftrightarrow a_k^* = a_k$

$\omega_0 = \frac{2\pi}{6} = \frac{\pi}{3}$

↳ Non-zero $a_k$'s can be $a_1, a_{-1}, a_2$ & $a_{-2}$

$$\boxed{a_{-1} = a_1^* \quad \& \quad a_{-2} = a_2^*}$$

④ $x(t) = -x(t-3)$

⑤ $\frac{1}{6}\int_{-3}^{3} |x(t)|^2 dt = \frac{1}{2} = \sum_k |a_k|^2 = 2|a_1|^2 +$

⑥ $a_1$ is real positive no. $\Rightarrow a_1 = a_{-1} =$ the real no.

$$x(t) = -x(t-3)$$

$$\underbrace{a_1 e^{j\omega_0 t} + a_1 e^{-j\omega_0 t}}_{\because a_{-1} = a_1} + a_2 e^{j2\omega_0 t} + a_2^* e^{-j2\omega_0 t} = -a_1 e^{j\omega_0(t-3)} - a_1 e^{-j\omega_0(t-3)} - a_2 e^{+j2\omega_0(t-3)} - a_2 e^{-j2\omega_0(t-3)}$$

$$\Rightarrow a_1 \underbrace{\left(e^{j\frac{\pi}{3}t} + e^{-j\frac{\pi}{3}t}\right)} + a_2 e^{j\frac{2\pi}{3}t} + a_2^* e^{-j\frac{2\pi}{3}t} = -a_1 \left(e^{j\frac{\pi}{3}t}\cdot e^{-j\pi} + e^{-j\frac{\pi}{3}t}\cdot e^{j\pi}\right) - a_2 e^{j\frac{2\pi}{3}t}\cdot e^{-j2\pi} - a_2^* e^{-j\frac{2\pi}{3}t}\cdot e^{j2\pi}$$

$$\Rightarrow \underbrace{a_1 e^{j\frac{\pi}{3}t} + a_1 e^{-j\frac{\pi}{3}t}}_{\hookrightarrow a_2 = a_{-2}^* = 0} + a_2 e^{j\frac{2\pi}{3}t} + a_2^* e^{-j\frac{2\pi}{3}t} = +a_1 e^{j\frac{\pi}{3}t} + a_1 e^{-j\frac{\pi}{3}t} - a_2 e^{j\frac{2\pi}{3}t} - a_2^* e^{-j\frac{2\pi}{3}t}$$

$a_1 = a_{-1} =$ +ve real coefficient

$$x(t) = a_1 e^{j\pi_3 t} + a_1 e^{-j\pi_3 t}$$

$$= 2a_1 \cos \pi_3 t$$

$$\therefore \frac{1}{6}\int_{-3}^{3} |x(t)|^2 dt = \frac{1}{2} = \sum_{k=-\infty}^{\infty} |a_k|^2$$

$$= 2|a_1|^2$$

$$|a_1|^2 = \frac{1}{4}$$

Since $a_1$ is +ve & real

$$\therefore a_1 = 1/2$$

$$\therefore x(t) = \cos \frac{\pi}{3} t = A \cos(Bt + C)$$

$$A = 1, \quad B = \pi_3 \quad \& C = 0$$

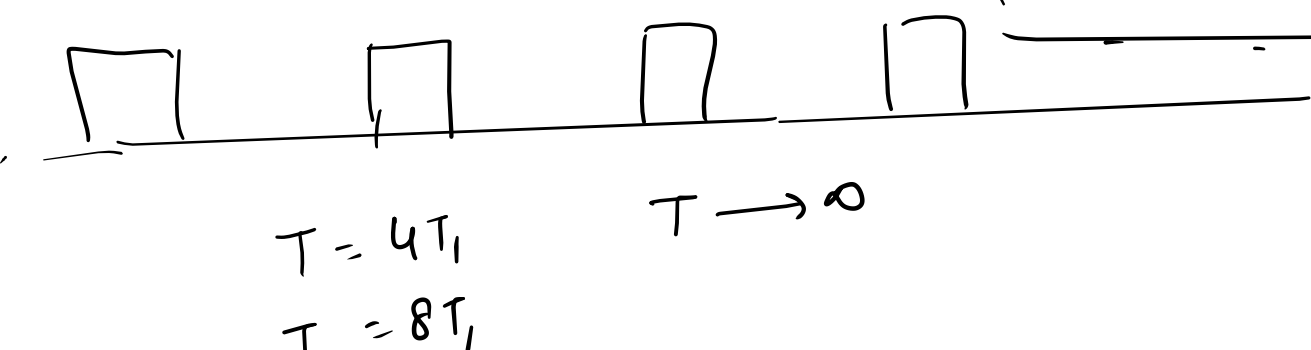
C.T. Signal $x(t)$

## C.T. Fourier Transform –

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt \Rightarrow \text{F. T.}$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega \Rightarrow \text{Inverse Fourier Transform}$$

$$x_p(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t}$$



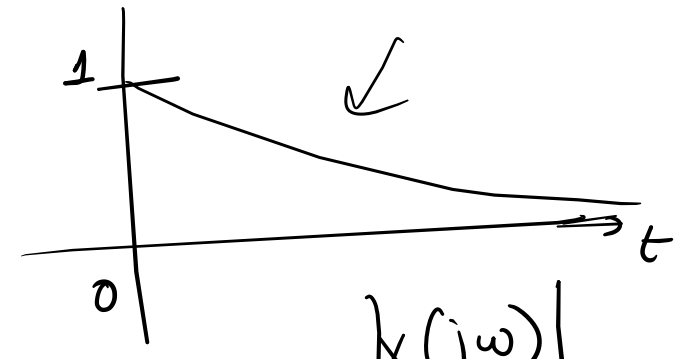
$T = 4T_1$

$T = 8T_1$

$T \to \infty$

ex1

$$x(t) = e^{-at}u(t) \qquad a>0$$




$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

$$H(s) = \int_{0^-}^{\infty} h(t) e^{-st}\, dt$$

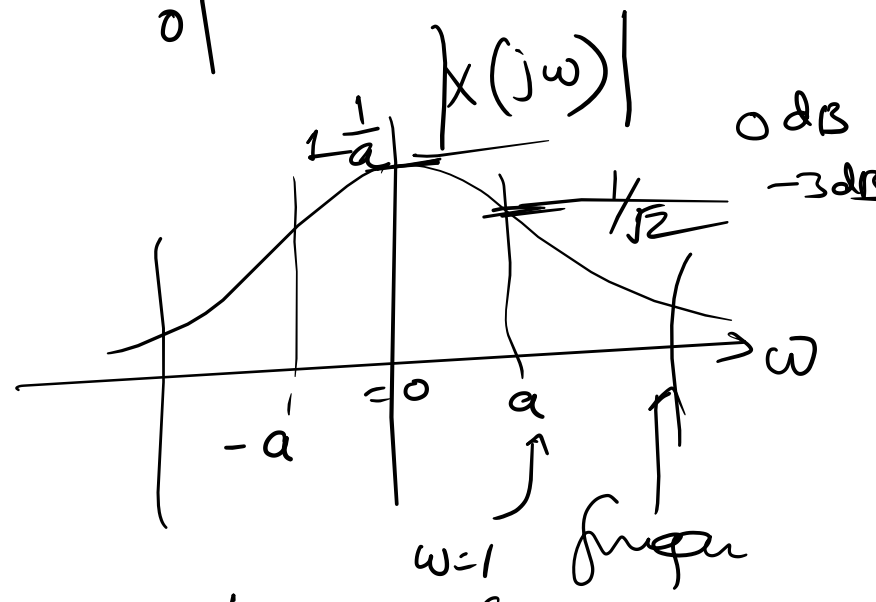
$$= \int_0^{\infty} e^{-at} \cdot e^{-j\omega t}\, dt = \int_0^{\infty} e^{-(a+j\omega)t}\, dt$$

$$\frac{e^{-(a+j\omega)t}}{-(a+j\omega)}\Bigg|_0^{\infty}$$

$$= \frac{1}{a+j\omega}\left[ e^{0}\cdot e^{j0} - e^{-\infty}\cdot e^{-j\infty} \right] = \frac{1}{a+j\omega}$$

$$= \frac{1}{a+j\omega}\left[ \underbrace{e^{0}\cdot e^{j0}}_{1} - \underbrace{e^{-\infty}\cdot e^{-j\infty}}_{0} \right]$$

$\angle X(j\omega) = -\tan^{-1}\omega/a$



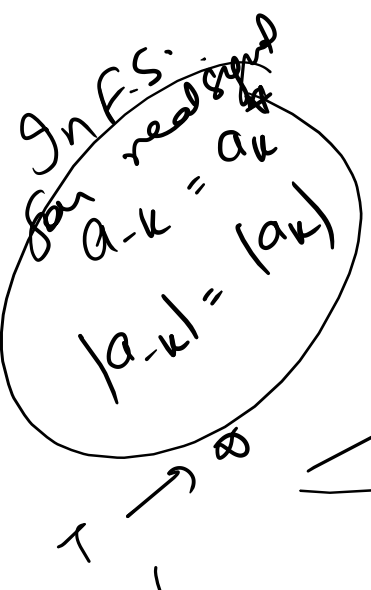

$$10\log_{10}|X(j\omega)|^2$$

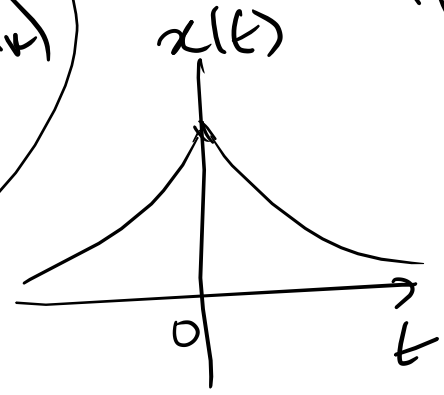
$$\frac{1}{1+j} = \frac{1}{\sqrt{2}}$$

Ex

$$x(t) = e^{-a|t|} \qquad a > 0 = \begin{cases} e^{-at} & t \geq 0 \\ e^{at} & t < 0 \end{cases}$$

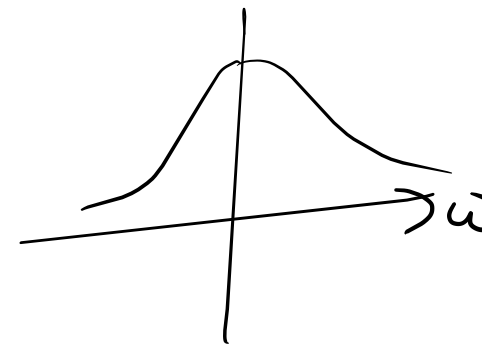
$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$




$$= \int_{-\infty}^{0} e^{at} e^{-j\omega t} dt + \int_{0}^{\infty} e^{-at} \cdot e^{-j\omega t} dt$$

$$X(j\omega) = \frac{1}{a - j\omega} + \frac{1}{a + j\omega} = \frac{2a}{a^2 + \omega^2}$$




If $x(t)$ is real, magnitude spectrum of $X(j\omega)$ is symmetric about y-axis or is an even fn.

If $x(t)$ is real & even, $X(j\omega)$ is also real & even.

$\delta(\omega)$

Ex

$$x(t) = \delta(t) :$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = 1$$

Time - frequency duality relationship $\Longrightarrow$

localized in time $\equiv$ infinite duration in frequency domain

$$\boxed{f(t) = 3} \quad \forall t \Rightarrow \boxed{F(j\omega) = 6\pi\delta(\omega)}$$

Ex

Inverse F.T. $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$

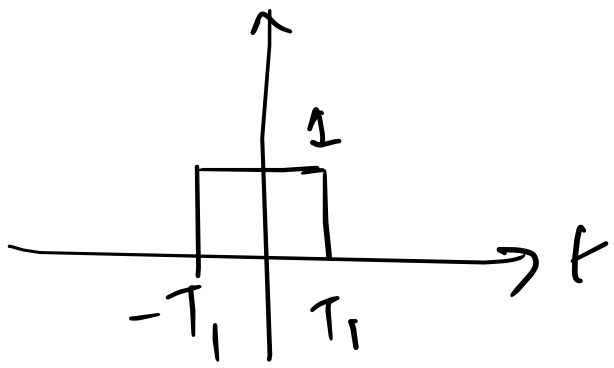
Let us assume $X(j\omega) = 3 . 2\pi \delta(\omega) \Rightarrow x(t) = 3 \int_{-\infty}^{\infty} \delta(\omega) e^{j\omega t} d\omega$

$= 3 . 1 \quad \forall t$

$= 3 \quad \forall t$

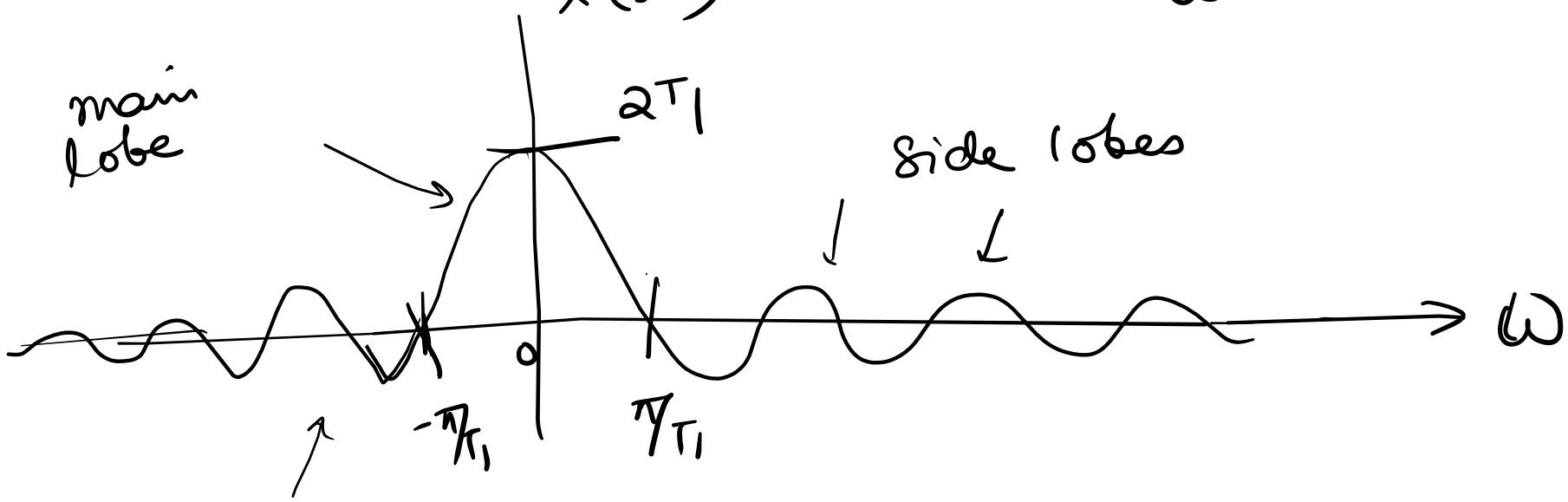
Ex

$$x(t) = \begin{cases} 1 & |t| < T_1 \\ 0 & |t| > T_1 \end{cases}$$



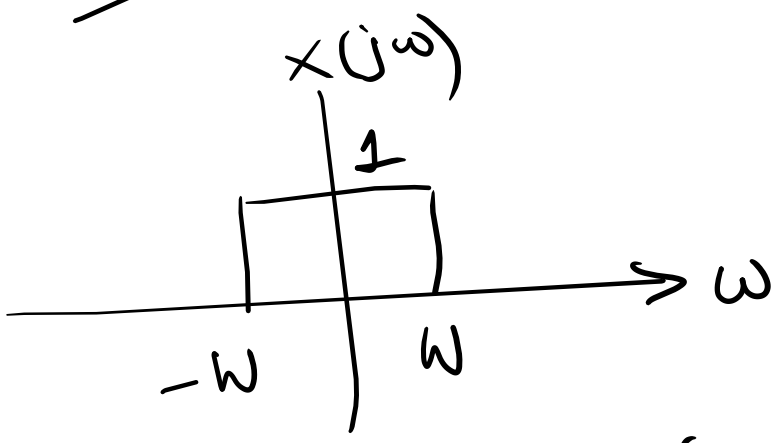

$$X(j\omega) = \int_{-T_1}^{T_1} e^{-j\omega t}\, dt = \frac{e^{-j\omega t}}{-j\omega}\Bigg|_{-T_1}^{T_1}$$

$$X(j\omega) = 2\,\frac{\sin \omega T_1}{\omega}$$



Ex

$$X(j\omega) = \begin{cases} 1 & |\omega| < W \\ 0 & \text{otherwise} \end{cases}$$




find $x(t)$ -

$$x(t) = \frac{1}{2\pi} \int_{-W}^{W} e^{j\omega t} d\omega$$

$$x(t) = \left| = \frac{1}{2\pi} \frac{e^{j\omega t}}{jt} \right|_{-W}^{W} = \frac{1}{\pi t} \left( \frac{e^{jWt} - e^{-jWt}}{2j} \right)$$

$$= \frac{\sin Wt}{\pi t}$$